

Effects of simvastatin and pravastatin on gonadal function in male hypercholesterolemic patients.

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Inhibition of cholesterol biosynthesis by hydroxymethyl glutaryl coenzyme A (HMG-CoA) reductase inhibitors could, in theory, adversely affect male gonadal function because cholesterol is a precursor of steroid hormones. The objective of this randomized double-blind trial was to compare the effects of simvastatin, pravastatin, and placebo on gonadal testosterone production and spermatogenesis. After a 6-week placebo and lipid-lowering diet run-in period, 159 male patients aged 21 to 55 years with type IIa or IIb hypercholesterolemia, low-density lipoprotein (LDL) cholesterol between 145 and 240 mg/dL, and normal basal levels of testosterone were randomly assigned to treatment with simvastatin 20 mg (n = 40), simvastatin 40 mg (n = 41), pravastatin 40 mg (n = 39), or placebo (n = 39) once daily. After 24 weeks of treatment, mean total cholesterol levels were decreased 24% to 27% and mean LDL cholesterol was decreased 30% to 34% in the 3 active-treatment groups ($P < .001$ for all comparisons to placebo). At 24 weeks, there were no statistically significant differences between the placebo group and any of the active-treatment groups for the change from baseline in testosterone, human chorionic gonadotropin (hCG)-stimulated testosterone, free testosterone index, follicle-stimulating hormone (FSH), luteinizing hormone (LH), or sex hormone-binding globulin (SHBG). Moreover, there were no statistically significant differences at week 12 or week 24 for the change from baseline in sperm concentration, ejaculate volume, or sperm motility for any active treatment relative to placebo. Both simvastatin and pravastatin were well tolerated. In summary, we found no evidence for clinically meaningful effects of simvastatin or pravastatin on gonadal testosterone production, testosterone reserve, or multiple parameters of semen quality.